

Meningitis - Summary

Key Symptoms: Classical triad: stiff neck, fever, altered mental status

- Other key symptoms: petechial rash, photophobia, Brudzinski/kernig's sign, low BP, seizures, etc.

Diagnosis: Cerebrospinal fluid culture/analysis (gold standard), blood cultures/lab values

Most common causing-pathogens: based on age

- **Neonates (<1 month):** Group B strep, E.coli and other G -ve bacilli, *Listeria monocytogenes*
- **1 month-23 months:** *S. pneumoniae*, *N. meningitidis*, *S. agalactiae*, *H. influenzae*, *E. coli*
- **2-50 years:** *N. meningitidis*, *S. pneumoniae*, *H. influenzae*
- **>50 years / Immunocompromised (alcohol use disorder, altered immune status, pregnancy):** *S. pneumoniae*, *N. meningitidis*, *L. monocytogenes*, G-ve bacilli

Pharmacological Treatment:

❖ Empiric therapy:

- **Neonates (<1 month):** Ampicillin + cefotaxime **OR** aminoglycoside
- **1 month-2 years, 2-50 years:** Cefotaxime **OR** ceftriaxone + vancomycin
- **>50 years or immunocompromised:** Ampicillin + (cefotaxime **OR** ceftriaxone) + vancomycin
- **Beta-lactam allergy:**
 - **1 month-2 years, 2-50 years:** Meropenem + vancomycin **OR** chloramphenicol + vancomycin (add TMP/SMX if immunocompromised)
 - **>50 years or immunocompromised:** TMP/SMX + (meropenem **OR** moxifloxacin) + vancomycin

❖ Targeted therapy:

- **Gram negative:** *H. influenzae*, *N. meningitidis*, *S. pneumoniae*
 - IV ampicillin (for beta-lactamase -ve *H. influenzae*) **OR** penicillin G (if susceptible for *N. meningitidis* or *S. pneumoniae*) **OR** 3rd gen cephalosporin (for all) for 7-14 days
 - **Beta-lactam allergy:** Cipro/levo/moxifloxacin, meropenem **OR** chloramphenicol (for *S. pneumoniae* also vancomycin + rifampin but not cipro)
- **Gram positive:** *S. agalactiae*, *L. monocytogenes*
 - Pen G **OR** ampicillin + gentamicin (for synergy) for 14-21 days (≥21 for *L. monocytogenes*)
 - **Beta-lactam allergy** [for *Listeria*]: TMP-SMX **OR** linezolid
- **Gram negative bacilli:**
 - 3rd gen cephalosporin (cefotaxime or ceftriaxone) +/- aminoglycoside (gentamicin or tobramycin) **OR** meropenem **OR** imipenem/cilastatin for 21 days
 - **Beta-lactam allergy:** Cipro/levo/moxifloxacin, meropenem **OR** chloramphenicol

Monitoring Parameters: Symptom resolution within 1-2 days after antibiotic treatment

Additional Key Points:

- Always add ampicillin in neonates, >50 and risk factors for *L. monocytogenes* coverage
 - If beta-lactam allergy, use TMP/SMX instead in these groups
- Adjunctive dexamethasone reduces cerebral edema, lowers intracranial pressure. Most beneficial when given before or at time of Abx administration
 - Data supportive in *adults* with *S. pneumoniae*, and *children* with *H. influenzae*